Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/25/2014
PI Name:	Douglas, Grace Ph.D.		
Project Title:	Bioavailability of Nutrients in Food		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	September 2014 final report: PI Changes: Monica Leong to Grace Douglas to Maya Cooper. Contributors: Emily Darchuk, Ryan Dowdy		
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
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Performance Goal Text:			
Task Description:	NASA is planning for long duration missions beyond low Earth orbit. In order to ensure sustainment of crew health and performance, adequate nutrition must be delivered by the food system and efficiently absorbed by crew members to prevent micronutrient deficiency. No collective source of literature on nutrient bioavailability in processed foods is available. The purpose of this task is to conduct a literature review of nutrient bioavailability taking into account the form of food ingredient interactions and processing conditions. The review will identify where information is inadequate and be used to propose formulation and processing suggestions for the development of food products with improved bioavailability. The Specific Aims are as follows: 1. Review current scientific literature on bioavailability of nutrients and how it is affected by form, ingredient		

	interactions and processing conditions. 2. Characterize the conditions that reduce or improve bioavailability of nutrients for future food development
Rationale for HRP Directed Research:	The review will contain both published and internal NASA documents on space food nutrient content-- information not accessible to the general public. Additionally, the processing variables which impact the nutritional quality of space food are well understood by the space food experts assigned to task.
Research Impact/Earth Benefits:	
Task Progress:	NASA is planning for long duration missions, and providing the necessary nutrition to ensure sustainment of crew health and performance is essential for success. Understanding the impact of processing and storage on the bioavailability of micronutrients as well as distinguishing between supplemental vitamin and natural vitamin forms is paramount to developing a food system to minimize risk of inadequate nutrition. Supplemental forms of vitamins and minerals generally are better absorbed by the body than the natural forms of the micronutrient though this phenomena is not always the case. The bioavailability of vitamin E and riboflavin is higher from natural sources as compared to their synthetic counterparts. Additionally, natural forms of vitamin C, biotin, calcium, magnesium, potassium, and zinc were shown to achieve equal serum concentrations as the supplements after consumption of like amounts in controlled test scenarios. Hence, as necessary, food fortification can occur with many of the micronutrients without adverse consequence to the physiological delivery provided that adequate stability of supplemental micronutrients can be achieved. Many ingredient interactions and matrix effects occur with micronutrients that impact nutrient absorption. One ingredient interaction which occurs during the absorption phase is the competitive absorption of similar compounds in the body. For example, beta-carotene competes with other carotenoids for intestinal absorption. Likewise, zinc limits iron absorption when the two are consumed simultaneously. Another interaction affecting bioavailability is the inhibitory nature of phytate and phytic acid, naturally occurring compounds in plants, with the mineral absorption of calcium, chromium, iron, magnesium, phosphorus, and zinc. Some vitamins, such as vitamins A and E, have increased bioavailability if some fat is consumed simultaneously. Vitamin A, riboflavin, and vitamin C show increased absorption if more dietary fiber is present at consumption so that the vitamins have a longer gastrointestinal residence time for absorption. Processing impacts to the bioavailability of micronutrients were very limited. Vitamin concentrations in foods are commonly affected by heat, water leaching, or oxidation. However only two vitamins actually had documentation of bioavailability differences that were dependent on the processing method. Vitamin A provitamins, because of their placement in the microstructure of plants, often become more bioavailable after cooking or pureeing. The niacin in grains must be released by acid or alkali extraction to allow significant absorption. Future work with regards to nutrient bioavailability in the space food system should focus on ensuring that micronutrients are provided in an optimal matrix within many food items. Both vitamin and mineral absorption in many instances was dose-dependent, meaning that percent absorption was higher at lower doses, up to a physiological nutrient saturation level, but overconsumption of micronutrients did not result in absorption at the same rate above saturation. Hence, micronutrients should be spread across many food items to ensure continued nutrient delivery at each meal occasion. Competing nutrients should be staggered in their consumption. Additionally, fortification efforts must take into account the data here in terms of surrounding food matrix and form of the supplemental micronutrient used.
Bibliography Type:	Description: (Last Updated: 11/22/2025)